SPM 5708 Homework 6 (Multiple Linear Regression)

6.1. Which Stats Matter in Football?

We've spent time looking at stats that are more or less relevant to the success of baseball teams. But what about football? Which basic, *team-level* stats have the largest effect on point margins? In this set of questions, we'll take an initial look.

1. Download the *Gridiron* dataset from Canvas. Import the *NFL* worksheet and keep it named *NFL*. Note that it contains basic team-level stats for all NFL teams from 2010-2017.

2. It has been said that efficiency stats are a better measure of performance than count stats. Test this by seeing if *passing yards per attempt* (*PassYPA*) has a stronger linear relationship with *points for* (*PF*) than total passing yards (*TotPassYds*) does with points for (*PF*). Report and interpret your results below.

   ANSWER: Below, using correlation, we can see that Passing yards per attempt has a stronger linear relationship with *points for* than total passing yards does with points.

```
> cor(NFL$PassYPA, NFL$PF)
[1] 0.7577183
> cor(NFL$TotPassYds, NFL$PF)
[1] 0.6402673
```

3. Now, do the same for *passing yards allowed per attempt* (*OPassYPA*) and *points against* (*PA*), and *total pass yards allowed* (*OTotPassYds*) and *points against* (*PA*). Report and interpret your findings.

   ANSWER: Below, using correlation, we can see that passing yards allowed per attempt has a stronger linear relationship with points against than total pass yards allowed does with points against.

```
> cor(NFL$OPassYPA, NFL$PA)
[1] 0.7116725
> cor(NFL$OTotPassYds, NFL$PA)
[1] 0.4632597
```

4. Create a new variable called *Margin* by subtracting *PA* from *PF*. This point-margin variable (*Margin*) will now serve as our dependent, *y*-variable.

5. Regress *Margin* on *PassYPA, OPassYPA, RushYPA, ORushYPA,* turnovers committed (*TOC*), turnovers forced (*TOF*), offensive penalty yards (*OffPenYds*), and defensive penalty yards (*DefPenYds*) in a multiple regression model. Type the equation you obtain for the model **and** interpret the coefficients below.

   ANSWER:

$$y = -33.9 + 67.9\,PassYPA - 59.3\,OPassYPA + 17.8\,RushYPA - 21.3\,ORushYPA - 4.5\,TOC + 4.5\,TOF - 0.09\,OffPenYds + 0.08\,DefPenYds$$

For every one unit change in x (PassYPA, OPassYPA, RushYPA, ORushYPA, TOC, TOF, OffPenYds, DefPenYds), y will change by: 67.9 or -59.3 or 17.8 or -21.3 or -4.5 or 4.5 or -0.09 or 0.08 respectively, holding the other x-variables constant.

6. What do these coefficients say about the importance of the rush vs. the importance of the pass? The importance of offense vs. defense?

   ANSWER: It seems that pass has a greater impact on y than rush does. As of offense vs defense, in general, it seems that when getting the absolute value for all offensive statistics they are in average greater than the defense statistics, except for ORushYPA which has a greater absolute value than RushYPA.

7. Did any of the resulting coefficients surprise you? If so, which ones and why?

   ANSWER: Offensive Rush Yards Per Attempt surprised me because of it having a higher impact than Rush Yards per Attempt. I wonder if that means that on average, a better defense is more important than trying to rush.

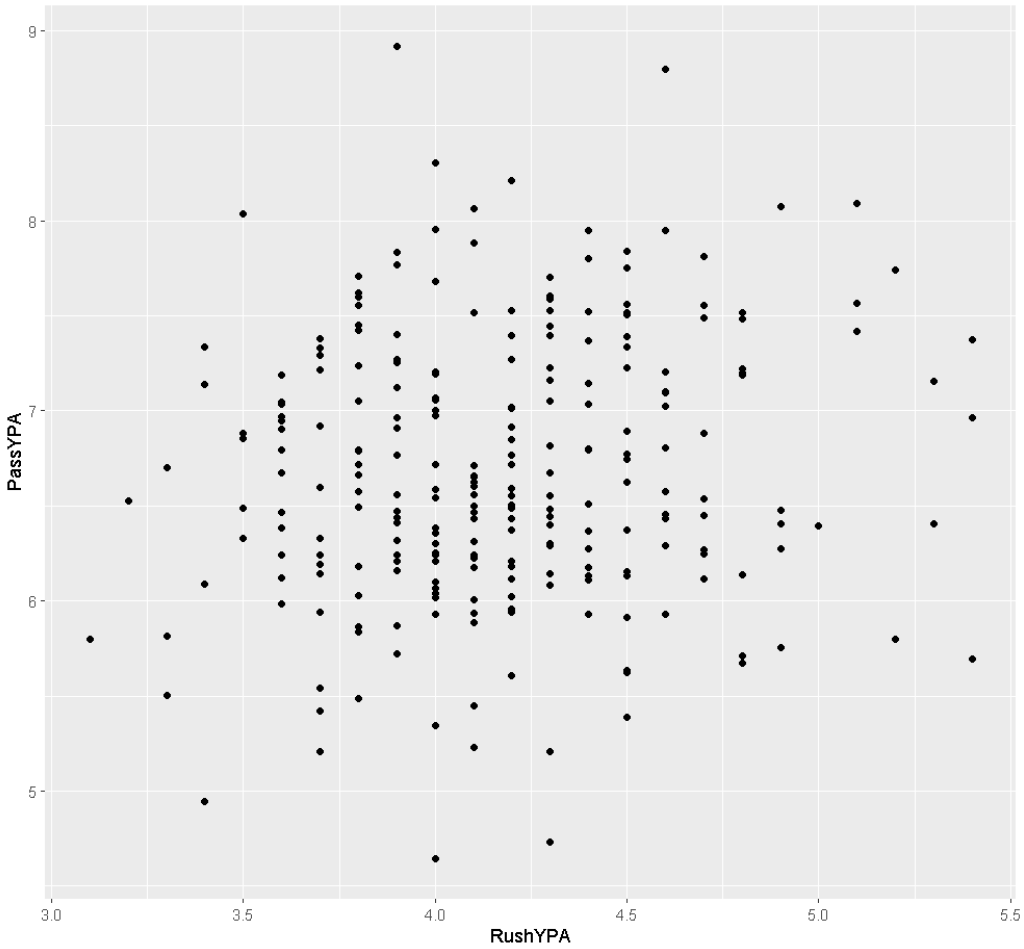
8. Look at the average salaries by position in the NFL. How do these seem to align with the results we just obtained?

   ANSWER: As we can see in the salaries by position website, offense leads with an average of 120.11 M while defense has an average of 83.22 M, so it seems that the average salaries align with the results we obtained.

9. It is often said that a good run offense is needed to set up a good pass offense. Test the strength between this relationship using the appropriate variables from the *Gridiron* data and create a scatterplot to visualize this relationship. Does there appear to be any relationship between efficient passing teams and efficient rushing teams?

   ANSWER: I decided to test the correlation between Rush Yards per attempt and Pass Yards per attempt.

```
> cor(NFL$RushYPA, NFL$PassYPA)
[1] 0.1251568
```



As we can see above, there is a rather a weak relationship or no relationship at all.

10. Import the *NFL2018* sheet into R. Using this new (test) data, generate predictions for each 2018 NFL team's point margin. Then, use these predictions to calculate the mean absolute error (MAE) for the model you created in question No. 5.

    ANSWER: For the model we created in question number 5, we get a mean absolute error of 49.56

11. Import the *NCAAF* sheet into R. Note that this sheet contains identical information for NCAA FBS teams between 2010 and 2017. Create an identical model to the one we used for the NFL data and type the resulting linear regression equation below.

    ANSWER:

$$y = -36.8 + 49\,PassYPA - 46.6\,OPassYPA + 46.8\,RushYPA - 58.5\,ORushYPA - 2.9\,TOC + 3.7\,TOF - 0.03\,OffPenYds + 0.13\,DefPenYds$$

12. How do the effects (coefficients) of these statistics vary in the NCAA compared to the NFL?

    ANSWER: I think one of the biggest changes in college football is the relevance of rushing. Passes seem to have slightly less weight on the Margin but now, rush has a bigger influence. Another big difference is that the team and the opponent's PassYPA and

OPassYPA seem to be closer in coefficient compared to the NFL coefficients for PassYPA and OPassYPA.

6.2. Estimating Attendance in Major League Baseball

Not every sport manager is a general manager. Many of you plan to work in other areas of the industry, such as marketing, operations, or event management. Thankfully, regression analysis can readily be applied to problems in these settings, too. For this set of questions, you will model the relationship between MLB attendance levels and a variety of quantitative factors believed to affect them.

1. Download the *MLB_Attendance* Excel file from Canvas and import the first worksheet labeled *MLB_Attendance*. This dataset contains game and attendance information from every regular-season game played during the 2014-2018 MLB seasons.

2. The *y*-variable of interest in this data is *PctCap*, a measure representing the percentage of the venue (stadium) capacity that was filled during a game. Why might we want to look at *PctCap* as the *y*-variable instead of the raw attendance figures (totals) recorded in the *Att* column?

   ANSWER: We choose PctCap because this allows to see the proportion of attendees compared to the capacity of each venue since every stadium is built differently and hence has a different capacity limit. So just raw attendance figures would not give an accurate representation of the relationship between MLB attendance levels and other factors.

3. Regress *PctCap* on the following *x*-variables recorded in the data: *GB_Pre*, *RS_PG*, *RA_PG*, *WPct_Pre*, *Pre_Streak*, *Temp*, *Wind*, *Start_Time*, *Opp_RS_PG*, *Opp_RA_PG*, *OppWPct_Pre*, and *Opp_Pre_Streak*. Run the *summary* function to view the results of the analysis (see the *Appendix* on the last page for *x*-variable descriptions).

```
lm(formula = PctCap ~ GB_Pre + RS_PG + RA_PG + WPct_Pre + Pre_Streak +
    Temp + Wind + Start_Time + Opp_RS_PG + Opp_RA_PG + OppWPct_Pre +
    Opp_Pre_Streak, data = MLB_Attendance)

Residuals:
     Min       1Q   Median       3Q      Max
-0.72358 -0.14427  0.01643  0.15442  0.65629

Coefficients:
                 Estimate Std. Error t value Pr(>|t|)
(Intercept)     7.305e-01  2.281e-02  32.025  < 2e-16 ***
GB_Pre          4.235e-03  2.427e-04  17.446  < 2e-16 ***
RS_PG           3.392e-02  3.933e-03   8.625  < 2e-16 ***
RA_PG          -5.673e-02  3.638e-03 -15.593  < 2e-16 ***
WPct_Pre       -1.268e-02  2.787e-02  -0.455   0.6492
Pre_Streak      5.352e-04  8.469e-04   0.632   0.5274
Temp            2.434e-03  1.737e-04  14.018  < 2e-16 ***
Wind            6.507e-03  3.577e-04  18.189  < 2e-16 ***
Start_Time     -1.701e-04  1.042e-05 -16.322  < 2e-16 ***
Opp_RS_PG       1.987e-02  3.892e-03   5.105 3.35e-07 ***
Opp_RA_PG      -3.151e-02  3.468e-03  -9.086  < 2e-16 ***
OppWPct_Pre    -5.286e-02  2.809e-02  -1.882   0.0599 .
Opp_Pre_Streak -8.504e-05  8.473e-04  -0.100   0.9201
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.1989 on 12225 degrees of freedom
Multiple R-squared:  0.1682,    Adjusted R-squared:  0.1674
F-statistic: 206.1 on 12 and 12225 DF,  p-value: < 2.2e-16
```

4. Does every *x*-variable have a significant effect on our measure of attendance (*PctCap*)? Which variables are not significant ($p > .05$)?

   ANSWER: No, the variables that are not significant are: WPct_Pre, Pre_Streak, OppWPct_Pre, and Opp_Pre_Streak.

5. Take the *x*-variable with the *highest p*-value and remove it from the regression equation. Then, re-run the model. Repeat this process (removing the non-significant *x*-variable with the highest *p*-value) until all remaining x-variables are significant ($p < .05$). Which x-variables were dropped from the model?

   ANSWER: The first variable dropped from the model was Opp_Pre_Streak. Then, WPct_Pre. Then, Pre_Streak. Then, OppWPct_Pre.

6. Type out the equation for your final model and interpret the coefficient (i.e., the effect of *x* on *y*) from at least two of the *x*-variables below.

   ANSWER:

$$y = 0.71 + 0.0042\,GB_Pre + 0.032\,RS_PG - 0.057\,RA_PG + 0.0024\,Temp + 0.0065\,Wind - 0.00017\,Start_Time + 0.014\,Opp_RS_PG - 0.027\,Opp_RA_PG$$

   For every 1 unit change in Temp, y (PctCap) will change by 0.0024, holding the other x variables constant. For every 1 unit change in Start_Time, y (PctCap) will change by -0.00017, holding the other x variables constant. For every 1 unit change in RS_PG, y (PctCap) will change by 0.032, holding the other x variables constant.

7. Report the model's R^2. Are you surprised at this value? Why or why not?

   ANSWER:

   - For our full model: 0.1682
   - For the model we got rid of Opp_Pre_Streak: 0.1682
   - For the model we got rid of Opp_Pre_Streak and Wpct_Pre: 0.1682
   - For the model we got rid of Opp_Pre_Streak, Wpct_Pre and Pre_Streak: 0.1682
   - For our final model where we got rid of Opp_Pre_Streak, Wpct_Pre, Pre_Streak and OppWPct_Pre: 0.1679

   I am surprised that the R^2 is not higher but it might just be that the x variables chosen in this case, even after getting rid of the insignificant ones, do not have a big effect on our response variable PctCap.

8. Now, import the *Att_2021* worksheet from the *MLB_Attendance* file. As the name implies, this contains game and attendance data from the 2021 MLB season. Use it to calculate your model's *mean absolute error (MAE)*. Report *and interpret* this value below.

   ANSWER: We get a mean absolute error equal to 0.2847573. Which means that on average, the model used was off by around 28%.

9. Which variables could potentially be added to improve this model? Do you think certain coefficients/effects might change if these variables were added?

   ANSWER: A variable that could be added to the model is Game, people might attend more games if the home team is having a winning season, so the number of the game matters. Another variable could be the day of the week, assigning a number to each day (1-7) and then see how it will affect the capacity percentage. I believe that adding variables that could be more relevant to the decision to come or not to a game could make a difference. Like for example, when I added Game to the model, the R^2 increased to 0.1809.

SUBMIT YOUR COMPLETED WORD DOC AND ASSOCIATED R SCRIPT TO THE CANVAS DROP BOX.

Appendix: 6.2. X-Variable Descriptions

GB_Pre: the number of games back the home team was from the division leader entering the game.

RS_PG: the average number of runs scored per game by the home team entering the game.

RA_PG: the average number of runs allowed per game by the home team entering the game.

WPct_Pre: the home team's winning percentage entering the game.

Pre_Streak: the winning (positive value) or losing (negative value) streak the home team was on entering the game.

Temp: the recorded temperature (in degrees Fahrenheit) at the time the first pitch was thrown in the observed game.

Wind: the average wind speed (in miles per hour) recorded at the venue during the game.

Start_Time: the number of minutes before (negative) or after (positive) noon that the game started.

Opp_RS_PG: the average number of runs scored per game by the visiting team entering the game.

Opp_RA_PG: the average number of runs allowed per game by the visiting team entering the game.

OppWPct_Pre: the visiting team's winning percentage entering the game.

Opp_Pre_Streak: the winning (positive value) or losing (negative value) streak the visiting team was on entering the game.